Fiscal Year:	FY 2018	**Task Last Updated:**	FY 07/24/2018
PI Name:	Ryder, Jeffrey Ph.D.		
Project Title:	Sweat Rates During Continuous and Interval Aerobic Exercise: Implications for NASA Multipurpose Crew Vehicle (MPCV) Missions		
Division Name:	Human Research		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:	HUMAN RESEARCH--Biomedical countermeasures		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	(1) **HHC**:Human Health Countermeasures		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	77058-3711	**Congressional District:**	36
Comments:	NOTE: Formerly at Universities Space Research Association.		
Project Type:	Ground	**Solicitation / Funding Source:**	2014-15 HERO NNJ14ZSA001N-Crew Health-OMNIBUS
Start Date:	10/01/2016	**End Date:**	03/31/2018
No. of Post Docs:	0	**No. of PhD Degrees:**	0
No. of PhD Candidates:	0	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA JSC
Contact Monitor:	Norsk, Peter	**Contact Phone:**	
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Flight Program:			
Flight Assignment:	NOTE: Period of performance changed to 10/01/2016-3/31/2018 per HRP element (original POP was 10/01/2015-9/30/2016 and then extended to 9/30/2017)--Ed., 8/3/17 NOTE: Extended to 9/29/2017, per HRP element ; original end date was 9/30/2016 (Ed., 1/31/17)		
Key Personnel Changes/Previous PI:	Juy 2018 report: Lori Ploutz-Snyder, Robert Ploutz-Snyder, and Jessica Scott are no longer CoIs on the project.		
COI Name (Institution):			
Grant/Contract No.:	Internal Project		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	Impaired crewmember performance due to loss of aerobic conditioning is one of the risks identified for mitigation by the NASA Human Research Program (HRP). Aerobic exercise within the small habitable volume of the Orion Multipurpose Crew Vehicle (MPCV) is expected to challenge the ability of the environmental control systems, especially in terms of moisture control. Exercising humans will contribute moisture to the cabin environment through sweating. Current acceptable moisture values are based on theoretical models that rely on an “average” crewmember working continuously at 75% of their aerobic capacity. Evidence suggests that high intensity interval exercise for shorter durations are effective in building and maintaining aerobic capacity. This investigation examines sweat rates for operationally relevant continuous and interval (4 x 4 minutes, 8 x 30 seconds) aerobic exercise protocols. The results will directly inform what types of aerobic exercise countermeasures will be feasible to prescribe for crewmembers aboard the MPCV. Results show that the 4 minute and 30 second interval protocols produce less sweat continuous exercise protocol. Furthermore, sweat production correlated with work performed on the ergometer and energy expenditure during exercise. These results suggest that interval exercise is an alternative to continuous aerobic exercise for situations where moisture production must be carefully monitored to avoid condensation issues.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	This research has direct operation impact to aid in exercise countermeasures prescription implementation for spaceflight missions where moisture removal is am operational constraint to continuous aerobic exercise.
Task Progress:	Thirty subjects completed five study sessions - two preliminary sessions and three experimental sessions. Subjects wore shorts and a t-shirt and were instructed to wear the same garments for all experimental trials. Trials were performed in an environmental chamber set to 23.9°C (75°F) and 34% relative humidity. No high-velocity fans were used to cool subjects. Aerobic capacity session - Aerobic capacity (VO2max) was determined by analysis of respiratory gases collected using a ParvoMedics True One 2400 Metabolic Measurements System (Sandy, UT). Rowing exercise is envisioned as the mode of exercise within the Orion cabin. Therefore, subjects were tested while exercising on a Concept2 Model E rowing ergometer (Morrisville, VT). An initial damper setting of 3 was used and adjusted according to subject preference and ability to achieve target workloads. Fixed workloads are not possible using a Concept2 rower, therefore subjects were given target workloads to try to maintain based on stroke-by-stroke power output measures displayed on the Concept2 Performance Manager 5 display. The initial workload target was 50 watts (W) for the first 3 minutes and increased by 25 W each minute until termination based on volitional fatigue or an inability to increase the power level for the next stage. For some individuals a “light” protocol was implemented. The protocol is similar to above, except the initial target load was 40 W and increased by 20 W every minute thereafter. Familiarization session - During the familiarization session, target workloads were determined for each of the three experimental sessions (continuous exercise, 4 minute intervals, and 30 second intervals). A linear relationship between watts and O2 utilization was used to estimate workloads corresponding to the 75% load to be used for continuous exercise. During familiarization, subjects rowed for 6 minutes at the calculated 75% VO2max load and oxygen consumption was measured. If the measured oxygen consumption was not within the 73-77% VO2max range, the target workload was adjusted accordingly. Following a 3 minute recovery at 50 W, subjects rowed for 4 minutes at 120% of the continuous target wattage. Workloads were adjusted as necessary to achieve a measured O2 consumption that reached 83-87% VO2max. After a 3 minute recovery period at 50 W, subjects performed two 30 second rowing sprints separated by 15 seconds of recovery. A guideline target wattage equal to 200% of the continuous exercise workload was used, but subjects were allowed to exceed the target if possible. Experimental sessions – Three aerobic training protocols were examined. All experimental sessions were conducted on a rowing ergometer as described above. The order of experimental sessions was randomly balanced for each order of implementation. These protocols have previously been implemented in HRP/NSBRI (National Space Biomedical Research Institute)-funded research in 14-day and 70-day bed rest studies (5-8) and International Space Station (ISS) spaceflight missions (NASA SPRINT study). Each protocol was followed by 30 minutes of seated resting recovery. Continuous protocol: Subjects rowed for at a target wattage corresponding to 75% of VO2max for 30 minutes. Thirty minutes of exercise was chosen because this is the longest amount of active time allotted for exercising crewmembers on MPCV missions. Four minute intervals: Subjects rowed for a 3 minute warm-up at 50 W followed by four 4 minute intervals (120% of continuous exercise target wattage, ~85% of VO2max at end of interval) each separated by 3 minutes of recovery at 50 W with a 2 minute 50 W cool down (30 total minutes). Thirty second intervals: Subjects perform a 3 minute warm up at 50 W, followed by eight 30 second all out intervals (~200% of continuous exercise target workload), each followed by 15 seconds of recovery at ~20 W of rest with a 2 minute cool down at a 50 W target (11 total minutes). Sweat loss - Subject nude dry body weight was recorded to the nearest 0.002 kg before exercise, following exercise, and thirty minutes following cessation of exercise using a Mettler Toledo platform scale and Mettler Toledo ICS 449 display unit (Columbus, OH) set to factory 5 second dynamic weight averaging. Sweat loss was calculated as the difference between weighings, with difference between water consumption weighing added to the change in body weight value [(body weightpre – body weightpost) + (water bottlepre – water bottlepost)](9). Sweat volume was assumed as a 1:1 ratio with mass. No subjects chose to drink water during the actual exercises. Subjects who elected to drink during the experiment all chose to drink following the post-exercise weighing. Subjects had a separate set of clothes and towel for the exercise and recovery periods. Pre-post weight of clothes and towels used to dry off prior to weighing was used to estimate the amount of water that was not immediately evaporated (i.e., liquid sweat). Power, work, heart rate - Stroke-by-stroke power and heart rate (using Polar H7 heart rate monitor, Polar USA, Bethpage, NY) were collected via the Concept2 Performance Manager paired by Bluetooth to the Concept2 ErgData application for iPad (Apple, inc., Cupertino, CA). Work was derived from power data over time. Metabolic measurements - Respiratory gas measurements were collected as described above using the ParvoMedics system. Continuous O2 uptake, CO2 production, respiratory exchange ratio, and energy expenditure were assessed

	during exercise, and as soon as logistically possible during the recovery period after the post-exercise weight was obtained. Total accumulated O2 and CO2 production was calculated for the exercise session. All O2 and CO2 data are expressed as standard temperature and pressure, dry gas. Results show that the 4 minute and 30 second interval protocols produce less sweat continuous exercise protocol. Furthermore, sweat production correlated with work performed on the ergometer and energy expenditure during exercise. These results suggest that interval exercise is an alternative to continuous aerobic exercise for situations where moisture production must be carefully monitored to avoid condensation issues.
Bibliography Type:	Description: (Last Updated: 08/11/2025)
Articles in Peer-reviewed Journals	Ryder JW, Crowell JB, Song HJ, Ewert M. "Sweat production during continuous and interval aerobic exercise." Aerosp Med Hum Perform. 2023 Aug;94(8):623-8. http://dx.doi.org/10.3357/AMHP.6171.2023 ; PMID: 37501297 , Aug-2023
Papers from Meeting Proceedings	Crowell JB, Ryder JW. "Sweat rates during continuous and interval aerobic exercise: Implications for NASA multipurpose crew vehicle missions." 2018 NASA Human Research Program Investigators' Workshop, Galveston, TX January 22-25, 2018. 2018 NASA Human Research Program Investigators' Workshop, Galveston, TX, January 22-25, 2018. , Jan-2018